

Supplementary Information

Radionuclide Identification for Plastic Scintillation Detectors by Backward Cumulative Channel Sum

Supplementary Table S1. Measurement conditions

Data ID	Category	Radionuclide(s)	Nominal activity (μCi)	Distance (cm)	Actual live time (s)	Role / note
Background	Background-only	-	-	-	20.000	Background correction and resampling
Sample-1	Single-source	Am-241	90	80	20.000	Am-241 reference spectrum
Sample-2	Single-source	Am-241	90	180	20.000	Single-source distance measurement
Sample-3	Single-source	Am-241	90	280	20.000	Single-source distance measurement
Sample-4	Single-source	Am-241	90	460	20.000	Single-source distance measurement
Sample-5	Single-source	Cs-137	16	70	19.284	Cs-137 distance series
Sample-6	Single-source	Cs-137	16	130	19.652	Cs-137 reference spectrum
Sample-7	Single-source	Cs-137	16	300	19.900	Cs-137 distance series
Sample-8	Single-source	Cs-137	16	390	19.962	Cs-137 distance series
Sample-9	Single-source	Co-60	9	60	18.672	Co-60 distance series
Sample-10	Single-source	Co-60	9	120	19.220	Co-60 reference spectrum
Sample-11	Single-source	Co-60	9	310	19.720	Co-60 distance series
Sample-12	Single-source	Co-60	9	470	19.864	Co-60 distance series
Sample-13	Mixed-source	Cs-137 + Co-60	16 / 9	136 / 136	19.134	Mixed-source dataset
Sample-14	Mixed-source	Am-241 + Co-60	90 / 9	287 / 287	19.676	Mixed-source dataset
Sample-15	Mixed-source	Cs-137 + Co-60	16 / 9	361 / 361	19.716	Mixed-source dataset
Sample-16	Mixed-source	Am-241 + Cs-137	90 / 16	163 / 163	19.628	Mixed-source dataset
Sample-17	Mixed-source	Am-241 + Cs-137 + Co-60	90 / 16 / 9	309 / 309 / 309	19.574	Mixed-source dataset
Sample-18	Mixed-source	Co-60 + Am-241	9 / 90	153 / 210	19.310	Separated source positions
Sample-19	Mixed-source	Cs-137 + Co-60	16 / 9	367 / 285	19.568	Separated source positions
Sample-20	Mixed-source	Am-241 + Cs-137	90 / 16	85 / 173	19.554	Separated source positions
Sample-21	Mixed-source	Cs-137 + Co-60	16 / 9	223 / 316	19.470	Separated source positions

Notes. Activities are nominal source activities. Distances are measured from the detector surface with each sealed point source aligned to the horizontal and vertical center of the detector panel. Samples 13-17 used the same distance for all constituents; Samples 18-21 used separated source positions. The nominal acquisition time was 20 s, and all calculations used the actual live time recorded in each MCA file. Channels 4-256 were treated as valid.

Supplementary Table S2a. Complete BCCS-NNLS results

Sample	Expected	Rnet (%)	Net cps	Am coefficient	Cs coefficient	Co coefficient	Residual (%)	Coefficient ranking
1	Am-241	23.53	1434.80	1.0000	0.0000	0.0000	0.00	Am-241 > Co-60 > Cs-137
2	Am-241	7.61	464.30	0.3157	0.0000	0.0054	5.78	Am-241 > Co-60 > Cs-137
3	Am-241	3.24	197.45	0.1288	0.0000	0.0000	26.57	Am-241 > Co-60 > Cs-137
4	Am-241	1.06	64.60	0.0423	0.0000	0.0000	78.98	Am-241 > Co-60 > Cs-137
5	Cs-137	132.28	8066.76	0.0000	2.5810	0.0000	1.35	Cs-137 > Co-60 > Am-241
6	Cs-137	52.00	3170.93	0.0000	1.0000	0.0000	0.00	Cs-137 > Co-60 > Am-241
7	Cs-137	12.62	769.69	0.0267	0.2197	0.0140	1.61	Cs-137 > Am-241 > Co-60
8	Cs-137	7.54	460.11	0.0551	0.1276	0.0000	6.61	Cs-137 > Am-241 > Co-60
9	Co-60	134.58	8206.81	0.0000	0.0000	2.7633	1.57	Co-60 > Cs-137 > Am-241
10	Co-60	49.50	3018.40	0.0000	0.0000	1.0000	0.00	Co-60 > Cs-137 > Am-241
11	Co-60	9.66	588.82	0.0282	0.0000	0.1827	1.35	Co-60 > Am-241 > Cs-137
12	Co-60	4.79	292.31	0.0237	0.0107	0.0745	3.84	Co-60 > Am-241 > Cs-137
13	Cs-137 + Co-60	89.72	5471.63	0.0323	0.9487	0.7973	0.48	Cs-137 > Co-60 > Am-241
14	Am-241 + Co-60	13.53	825.31	0.1405	0.0128	0.1954	1.31	Co-60 > Am-241 > Cs-137
15	Cs-137 + Co-60	16.85	1027.84	0.0606	0.1771	0.1284	1.20	Cs-137 > Co-60 > Am-241
16	Am-241 + Cs-137	44.56	2717.22	0.3575	0.6898	0.0050	0.53	Cs-137 > Am-241 > Co-60
17	Am-241 + Cs-137 + Co-60	24.68	1505.00	0.1718	0.2244	0.1879	1.31	Cs-137 > Co-60 > Am-241
18	Co-60 + Am-241	38.23	2331.37	0.2423	0.0000	0.6548	0.78	Co-60 > Am-241 > Cs-137
19	Cs-137 + Co-60	20.45	1247.37	0.0582	0.1636	0.2175	1.12	Co-60 > Cs-137 > Am-241
20	Am-241 + Cs-137	54.82	3343.30	0.9461	0.6093	0.0191	0.87	Am-241 > Cs-137 > Co-60
21	Cs-137 + Co-60	29.65	1807.91	0.0756	0.2129	0.3418	1.30	Co-60 > Cs-137 > Am-241

Notes. Reference rows are shaded. Coefficients are detector-response scaling factors relative to Samples 1, 6, and 10; they do not represent activity fractions.

Supplementary Table S2b. Raw-spectrum and five-window NNLS results

Sample	Raw bg	Raw Am	Raw Cs	Raw Co	Raw residual (%)	Raw ranking	EW Am	EW Cs	EW Co	EW residual (%)	EW ranking
1	1.0000	1.0000	0.0000	0.0000	0.00	Am-241 > Co-60 > Cs-137	1.0000	0.0000	0.0000	0.00	Am-241 > Co-60 > Cs-137
2	0.9667	0.3782	0.0182	0.0201	0.94	Am-241 > Co-60 > Cs-137	0.3154	0.0000	0.0048	1.10	Am-241 > Co-60 > Cs-137
3	0.9986	0.1561	0.0000	0.0000	0.99	Am-241 > Co-60 > Cs-137	0.1490	0.0000	0.0000	5.68	Am-241 > Co-60 > Cs-137
4	0.9790	0.0941	0.0186	0.0000	1.42	Am-241 > Cs-137 > Co-60	0.0607	0.0000	0.0000	14.72	Am-241 > Co-60 > Cs-137
5	0.9855	0.0000	2.5394	0.0000	3.05	Cs-137 > Co-60 > Am-241	0.0000	2.5582	0.0000	4.32	Cs-137 > Co-60 > Am-241
6	1.0000	0.0000	1.0000	0.0000	0.00	Cs-137 > Co-60 > Am-241	0.0000	1.0000	0.0000	0.00	Cs-137 > Am-241 > Co-60
7	0.9868	0.0393	0.2329	0.0195	1.11	Cs-137 > Am-241 > Co-60	0.0228	0.2192	0.0121	2.63	Cs-137 > Am-241 > Co-60
8	0.9754	0.0578	0.1665	0.0047	0.97	Cs-137 > Am-241 > Co-60	0.0226	0.1401	0.0000	8.01	Cs-137 > Am-241 > Co-60
9	1.0181	0.0000	0.0000	2.6037	3.80	Co-60 > Cs-137 > Am-241	0.0000	0.0000	2.7476	4.27	Co-60 > Cs-137 > Am-241
10	1.0000	0.0000	0.0000	1.0000	0.00	Co-60 > Am-241 > Cs-137	0.0000	0.0000	1.0000	0.00	Co-60 > Cs-137 > Am-241
11	0.9827	0.0591	0.0108	0.1894	0.95	Co-60 > Am-241 > Cs-137	0.0246	0.0021	0.1802	1.45	Co-60 > Am-241 > Cs-137
12	0.9884	0.0483	0.0164	0.0791	0.99	Co-60 > Am-241 > Cs-137	0.0338	0.0034	0.0764	3.07	Co-60 > Am-241 > Cs-137
13	1.0255	0.0000	0.9353	0.7672	1.60	Cs-137 > Co-60 > Am-241	0.0270	0.9653	0.7794	1.12	Cs-137 > Co-60 > Am-241
14	0.9592	0.2175	0.0312	0.2183	0.88	Co-60 > Am-241 > Cs-137	0.1449	0.0025	0.2022	2.57	Co-60 > Am-241 > Cs-137
15	0.9869	0.0707	0.1907	0.1347	0.99	Cs-137 > Co-60 > Am-241	0.0568	0.1740	0.1297	1.02	Cs-137 > Co-60 > Am-241
16	1.0092	0.3412	0.6829	0.0029	0.97	Cs-137 > Am-241 > Co-60	0.3591	0.6922	0.0000	1.43	Cs-137 > Am-241 > Co-60
17	0.9623	0.2088	0.2631	0.2011	1.16	Cs-137 > Am-241 > Co-60	0.1538	0.2274	0.1877	2.41	Cs-137 > Co-60 > Am-241
18	0.9999	0.2550	0.0043	0.6387	1.11	Co-60 > Am-241 > Cs-137	0.2512	0.0013	0.6475	1.46	Co-60 > Am-241 > Cs-137
19	0.9658	0.1074	0.1894	0.2317	0.96	Co-60 > Cs-137 > Am-241	0.0475	0.1658	0.2147	1.14	Co-60 > Cs-137 > Am-241
20	1.0302	0.8680	0.6046	0.0043	1.15	Am-241 > Cs-137 > Co-60	0.9354	0.6221	0.0069	0.46	Am-241 > Cs-137 > Co-60
21	0.9840	0.0883	0.2432	0.3279	1.11	Co-60 > Cs-137 > Am-241	0.0560	0.2315	0.3265	1.62	Co-60 > Cs-137 > Am-241

Notes. Raw bg is the fitted coefficient of the measured background template. Raw-spectrum and energy-window residuals are calculated in different representation spaces and should not be treated as identical statistical quantities.

Supplementary Table S3. Parametric Poisson resampling

Sample	Expected	Am median	Am 95% interval	Cs median	Cs 95% interval	Co median	Co 95% interval	Ranking stability (%)	All positive (%)	Rnet (%)	BCCS ϵ (%)
2	Am-241	0.3115	0.2784-0.3416	0.0000	0.0000-0.0116	0.0040	0.0000-0.0169	100.00	-	7.61	5.78
3	Am-241	0.1264	0.0845-0.1608	0.0000	0.0000-0.0041	0.0000	0.0000-0.0074	100.00	-	3.24	26.57
4	Am-241	0.0413	0.0000-0.0784	0.0000	0.0000-0.0032	0.0000	0.0000-0.0043	96.86	-	1.06	78.98
5	Cs-137	0.0000	0.0000-0.0000	2.5806	2.5611-2.6017	0.0000	0.0000-0.0000	100.00	-	132.28	1.35
7	Cs-137	0.0271	0.0000-0.0564	0.2196	0.2002-0.2367	0.0139	0.0000-0.0323	100.00	-	12.62	1.61
8	Cs-137	0.0550	0.0200-0.0911	0.1275	0.1109-0.1433	0.0000	0.0000-0.0043	99.84	-	7.54	6.61
9	Co-60	0.0000	0.0000-0.0000	0.0000	0.0000-0.0000	2.7635	2.7440-2.7824	100.00	-	134.58	1.57
11	Co-60	0.0231	0.0000-0.0504	0.0000	0.0000-0.0193	0.1809	0.1649-0.1943	100.00	-	9.66	1.35
12	Co-60	0.0226	0.0000-0.0516	0.0106	0.0000-0.0300	0.0743	0.0570-0.0903	99.84	-	4.79	3.84
13	Cs-137 + Co-60	0.0321	0.0000-0.0657	0.9485	0.9238-0.9739	0.7978	0.7746-0.8195	100.00	-	89.72	0.48
14	Am-241 + Co-60	0.1399	0.1117-0.1690	0.0129	0.0000-0.0320	0.1953	0.1769-0.2121	100.00	-	13.53	1.31
15	Cs-137 + Co-60	0.0608	0.0312-0.0900	0.1772	0.1566-0.1977	0.1282	0.1091-0.1468	100.00	-	16.85	1.20
16	Am-241 + Cs-137	0.3601	0.3277-0.3947	0.6874	0.6685-0.7038	0.0050	0.0000-0.0233	100.00	-	44.56	0.53
17	Am-241 + Cs-137 + Co-60	0.1719	0.1422-0.2020	0.2243	0.2031-0.2446	0.1881	0.1687-0.2077	-	100.00	24.68	1.31
18	Co-60 + Am-241	0.2379	0.2056-0.2687	0.0000	0.0000-0.0194	0.6531	0.6346-0.6683	100.00	-	38.23	0.78
19	Cs-137 + Co-60	0.0582	0.0286-0.0882	0.1637	0.1432-0.1842	0.2177	0.1984-0.2373	100.00	-	20.45	1.12
20	Am-241 + Cs-137	0.9468	0.9134-0.9790	0.6093	0.5878-0.6293	0.0193	0.0000-0.0380	100.00	-	54.82	0.87
21	Cs-137 + Co-60	0.0756	0.0448-0.1052	0.2131	0.1918-0.2347	0.3418	0.3218-0.3613	100.00	-	29.65	1.30

Notes. Each row summarizes 5,000 Poisson resamples of the measurement and background spectra with the measured reference matrix held fixed. The empirical 99th-percentile background-only maximum-coefficient gate was 0.027. The intervals do not include reference acquisition, positioning, calibration, gain, temperature, or environmental-background uncertainty.

Supplementary Table S4. Five-window reference responses

Reference	W1 net cps	W2 net cps	W3 net cps	W4 net cps	W5 net cps	Reference condition
Am-241	1389.550	43.700	3.150	7.050	-8.650	Sample-1; 80 cm; 90 μ Ci
Cs-137	883.574	1364.590	656.126	239.686	26.960	Sample-6; 130 cm; 16 μ Ci
Co-60	408.499	731.173	548.195	1070.043	260.491	Sample-10; 120 cm; 9 μ Ci